

Telco Customer Churn Analytics

Welcome to the Remote Work & Freelancing Technical Training Program – **Data Analysis Track**

This program is under **the Youth Grow Activity**, funded by the **U.S. Government**, implemented by **Q2 Impact** and **Partners for Good**, and executed by **parachute16**.

Case Scenario:

You have been hired as a **Junior Data Analyst** in the Customer Experience team of a telecommunications company.

Your mission is to use SQL to explore, clean, analyze, and prepare data from the **Telco Customer Churn database** [[Link](#)] to help the company:

- Understand patterns of customer churn
- Identify high-risk customer segments
- Prepare a clean reporting dataset for Power BI

You must demonstrate all skills learned throughout the course: queries, joins, aggregation, subqueries, schema creation, DDL/DML, views, functions, and exporting output.

PHASE 1: Database Setup & Documentation

Tasks

1. Create a new database in SQL Server called TelcoChurnAnalysis. 2. Create the Telco tables using DDL (CREATE TABLE), based on the Kaggle dataset: •

Customers

- Services (Internet/Phone/Streaming/etc.)
- Contracts / Billing
- Churn labels

3. Apply:

- Appropriate data types (INT, VARCHAR, DATE)
- Constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, DEFAULT

4. Insert data using INSERT INTO or BULK INSERT (if CSV import is used).

5. Write a 1-paragraph explanation of:

- Why databases matter in data analysis
- Difference between spreadsheets and databases
- Why this dataset is relational

Deliverable (Submit as PDF or DOCX)

- SQL scripts: CREATE TABLE, ALTER, INSERT
- Screenshot showing all tables created in SQL Server
- Short written explanation

PHASE 2: Exploratory SQL Queries

Tasks

1. Retrieve the following:

- All customer IDs and their tenure
- Customers with charges above \$100
- Customers who use fiber optic AND have monthly contract

2. Show:

- Sorted list of customers by tenure (descending)
- Top 15 customers with highest total charges

3. Use aliases to enhance readability:

- customer_id AS CustomerID
- monthly_charges AS MonthlyFee

4. Write queries demonstrating correct **logical query order**.

Deliverables

→ SQL script file: phase2_queries.sql →



Output screenshots for 3 selected queries

PHASE 3: Data Manipulation, Querying & Schema Maintenance

3.1 DDL: Data Definition Language

Tasks

1. Create missing tables or adjust structure using:
 - CREATE TABLE
 - ALTER TABLE
 - DROP TABLE (Explain why dangerous)
2. Apply essential constraints:
 - PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, DEFAULT
3. Modify data types where necessary (e.g., converting inconsistent fields to INT or DATE).

3.2 DML: Data Manipulation Language

Tasks

1. Insert new records into reference tables (e.g., payment types, contract types) using INSERT.
2. Update inconsistent or missing values in customer tables using UPDATE.
3. Delete test records or duplicates using DELETE.
4. Compare **TRUNCATE** vs **DELETE** and explain why TRUNCATE is not used on production tables.

3.3 DQL: Data Query Language

Tasks

1. Structure Validation Queries

- Retrieve all table names in the database using `INFORMATION_SCHEMA.TABLES`.
- Retrieve all columns and their data types from 2 tables using `INFORMATION_SCHEMA.COLUMNS`.



- Confirm constraints exist (PRIMARY KEY, FOREIGN KEY) using metadata queries.

2. Data Quality Validation

- Query to check for:
 - Null values in critical columns (e.g., customerID, contract type).
 - Inconsistent values (example: negative charges, incorrect text categories).
- Query showing the count of distinct values for:
 - Contract types
 - Payment methods
 - Internet service types

3. Post-cleaning Verification Queries

- Retrieve customers updated in PHASE 3 using `WHERE last_updated >= GETDATE()`
- List records changed or removed (soft delete example if used).
- Verify updated data types by running SELECT expressions using:
 - `ISNUMERIC()`
 - `TRY_CAST()`

4. Exploratory DQL for Schema Validation

- Retrieve first 20 records using `TOP (20)` to visually inspect cleaned data.
- Sort by fields most affected by DML (e.g., monthly charges, total charges).
- Aliases required for readability.

Deliverables

SQL Deliverables

- phase3_ddl.sql (CREATE, ALTER, DROP)
- phase3_dml.sql (INSERT, UPDATE, DELETE)
- phase3_dql.sql (Structure validation + Data quality checks + Verification queries)

Documentation Deliverables

1-2 pages explaining:

- When to use DDL vs DML vs DQL
- Why metadata queries are important in data validation



- Real-world scenarios where each component is used (e.g., schema update, cleaning pipeline)

PHASE 4: Aggregation, Grouping & Churn Insights

Tasks

1. Write queries to find:
 - A. Total number of customers
 - B. Average monthly charges by contract type
 - C. Number of churned vs non-churned customers
 - D. Top Contract Type with highest churn rate
 - E. Average tenure for customers grouped by payment method
 - F. Use HAVING to filter groups:
 - Payment methods with more than 1,000 customers
2. Ensure:
 - Use of COUNT, SUM, AVG, MIN, MAX
 - NULL handling using COALESCE or ISNULL

Deliverables

- SQL script: phase4_aggregations.sql
- A short summary of churn insights (5-7 bullet points)

PHASE 5: Joins & Multi-table Analysis

Tasks

1. Using INNER JOIN, list customers with their contract and monthly charge info.
 2. Using LEFT JOIN, show customers who have internet service vs those who don't.
 3. Using RIGHT JOIN, compare payment methods with customer info.
 4. Using FULL OUTER JOIN, find unmatched records between customer table and billing table.
 5. Use a SELF JOIN to identify customers with the same tenure.
 6. Use a 3-table join to analyze:
 - Customer
- Contract type
 - Internet service

Tasks

1. Write a scalar subquery to find:

Deliverables

→ SQL script: phase5_joins.sql



PHASE 6: Subqueries & CTEs

- Customers whose monthly charges are above the average monthly charge.
2. Write a table subquery returning top 10 high-churn-risk customers (based on charges + tenure + service type).
 3. Write a correlated subquery to find customers whose total charges exceed the average of their payment method group.
 4. Build two CTEs:
 - CTE1: Clean dataset (replace NULLs, remove invalid records)
 - CTE2: Churn segmentation (tenure categories: 0-12, 13-24, 25+)

5. Compare when CTE is better than a subquery (write 3 lines).

Deliverables

→ SQL script: phase6_subqueries_ctes.sql

PHASE 7: Built-In Functions

Tasks

1. Use CONCAT to build a full customer name (if applicable).
2. Use UPPER/LOWER to standardize categorical fields.
3. Calculate customer age using DATEDIFF (if date of birth exists).
4. Create a derived column:

```
customer_risk_level = CASE WHEN tenure < 12 AND churn='Yes' THEN 'High Risk'  
ELSE 'Low/Medium' END
```



5. Convert total charges to integer using CAST/CONVERT.

Deliverables

→ SQL script: phase7_functions.sql

PHASE 8: Views for Reporting

Tasks

Create three (3) views:

1. vw_customer_overview
 - Basic customer profile fields
2. vw_churn_summary
 - Contract type
 - Tenure category
 - Monthly charges
 - Whether customer churned

3. vw_high_risk_customers

- Customers flagged as high churn risk (from CASE statement earlier)

Explain:

- When to use Views vs CTEs
- Benefits of views for reporting

Deliverables

- SQL script: phase8_views.sql
- Brief explanation (5–6 lines)

PHASE 9: Reporting Output & Data Export

Tasks

aliases.

1. Write a final query combining:

- Customer info

3. Export the result from SQL Server to: •

CSV

- Excel (.xlsx)

- Contract details

- Internet service

- Churn flag

- Calculated fields (risk, tenure category) 2.

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Format column names cleanly using

4. **(Optional)** Connect SQL Server to Power BI and load the dataset.

Deliverables

- CSV export
- Excel export
- Final query script
- Screenshot from Power BI (optional)

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FINAL SUBMISSION PACKAGE

1. SQL Scripts

- phase1_setup.sql
- phase2_queries.sql
- phase3_ddl_dml_dql.sql
- phase4_aggregations.sql
- phase5_joins.sql
- phase6_subqueries_ctes.sql
- phase7_functions.sql
- phase8_views.sql
- final_reporting_query.sql

2. Documentation

- Written explanations (Phase 1, 3, 6, 8)
- Insight summary (Phase 4)

3. Exports

- Final CSV
- Final Excel file

4. Screenshots

- SQL Server database schema
- Query outputs where required
- **Optional** Power BI dashboard